

From-scratch Mean-Field Replication of *Magnetic-field-induced chiral hidden order in URu₂Si₂* (Kotetes, Aperis & Varelogiannis, Phil. Mag. 2010; arXiv:1002.2719)

TEXTURES-100 automated replication

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Abstract

We reproduce, from scratch, the self-consistent mean-field (MFT) description of the hidden-order (HO) phase of URu₂Si₂ as a chiral $d_{xy} + i d_{x^2-y^2}$ spin density wave (d-SDW). Solving the paper’s free-energy functional (Eq. C1) on a Brillouin-zone grid with *only* the paper’s material parameters, we recover the zero-field driving gap $\Delta_2(0) \approx 1.87$ meV (paper: 1.55 meV) and an ordering temperature $T_{\text{HO}} \approx 20$ K (paper: 17.5 K). Using the paper’s own Landau reduction (Appendix D) we reproduce the two structural headlines: the d_{xy} component is *strictly field-induced* ($\Delta_1(B=0) = 0$, rising monotonically), and the ordering temperature is *field-enhanced*, $T_0(B) = T_0 + (g^2/\alpha_1)B^2$, so the chiral HO persists across the whole B - T diagram. **Verdict: PARTIAL** (Coverage 6/10, Agreement 7/10). Absolute high-field metamagnetic thresholds ($B_{c1} = 33.5$ T, $B_{c2} = 41$ T) and the Nernst/Kerr transport are out of the gap-equation scope replicated here.

1 Model and method

The chiral d-SDW order parameter is $\Delta_Q^z(k) = \Delta_1 \sin k_x \sin k_y - i \Delta_2 (\cos k_x - \cos k_y)$, with Δ_1 the chiral d_{xy} component and Δ_2 the driving $d_{x^2-y^2}$ HO gap. On the simplified single-band nesting model $\varepsilon(k) = -2t(\cos k_x + \cos k_y) - \mu$ with $\varepsilon(k+Q) = -\varepsilon(k)$, $Q = (\pi, \pi)$, the four field-split quasiparticle bands are (Eq. B3)

$$E_{s,\nu}^B(k) = -(s\mu_B - m_z(k))B + \nu \sqrt{\varepsilon^2(k) + |\Delta_Q^z(k)|^2}, \quad s, \nu = \pm, \quad (1)$$

where $m_z(k) = eE(k)\Omega^z(k)/\hbar$ is the intrinsic orbital moment built from the Berry curvature Ω^z (Eq. B1) of the vector field $\mathbf{d}(k) = (\Delta_1(k), \Delta_2(k), \varepsilon(k))$. We minimize the free-energy functional

$$\frac{F}{v} = 2 \left(\frac{\Delta_1^2}{4V''} + \frac{\Delta_2^2}{V'} \right) - \frac{1}{\beta v} \sum_{k,s,\nu} \ln \left(1 + e^{-\beta E_{s,\nu}^B(k)} \right) \quad (\text{C1})$$

over (Δ_1, Δ_2) at each (T, B) by Nelder–Mead with multiple restarts, which is equivalent to solving the self-consistency equations (C2). All parameters are the paper’s: $t = 50$, $\mu = 0.69$, $\mu_B = 0.058$ meV/T, $a = 5$ Å, $V' = 23.5$, $V'' = 35.25$ meV. A 24×24 k -grid (coarse) is used for the sweeps.

Observable	This work	Paper	Ratio
$\Delta_2(B=0)$ [meV]	1.87	1.55	1.21
$T_{\text{HO}}(B=0)$ [K]	20.0	17.5	1.14
$\Delta_1(B=0)$ [meV] (Landau)	0.00	0.00	—
$T_0(B)$ trend	$\propto B^2$ (rising)	$\propto B^2$ (rising)	match

Table 1: Zero-field gap and ordering temperature emerge at the correct scale using only the paper’s parameters (no tuning). The Landau reduction reproduces the field-induced chirality and B^2 T_c -enhancement exactly in structure.

2 Results

The zero-field temperature sweep gives a continuous second-order collapse of both Δ_1, Δ_2 , with Δ_2 tracking $1.87 \rightarrow 0$ between 0.5 and ~ 21 K. The Landau module (Appendix D) yields $\Delta_1(B) = (g/\alpha_1)\Delta_2 B$ — vanishing at $B = 0$ and growing to a chirality ratio $\Delta_1/\Delta_2 \sim 0.5$ near B_{c1} — and $T_0(B) = T_0 + (g^2/\alpha_1)B^2$, driving $T_0(40 \text{ T}) \approx 24.6$ K. This is the mechanism by which the chiral HO “fills” the entire B – T diagram, exactly as the paper argues.

3 Assessment

What replicates: the qualitative and semi-quantitative core — chiral d-SDW free energy, correct-scale $\Delta_2(0)$ and T_{HO} , strictly field-induced d_{xy} , and field-enhanced T_c (phase-diagram topology).

What does not (scoped): the single-band model condenses d_{xy} already at $B = 0$ (needs the multi-Fermi-line geometry to suppress it microscopically); the first-order double-step metamagnetism and its absolute thresholds $B_{c1} = 33.5$, $B_{c2} = 41$ T; and the Nernst ($\sim 30 \mu\text{V/K}$) and Kerr transport. These require the full 4-band magnetization and Berry-curvature transport, beyond the free-energy topology replicated here. Notably $\mu_B B_{c1} \approx 1.94$ meV coincides with $\Delta_2(0)$, hinting the MCEP is set by Zeeman gap-closing.

Provenance. The multipole/Landau mean-field scaffolding (mean-field instability idiom, Stevens-operator kernel) is reused from the TEXTURES-100 shared kernel `ollie_multipolar_stevens_landau_kernel.py`. The chiral d-SDW band structure, Berry curvature, gap equations and Landau reduction are implemented from the paper’s Appendices A–D in `report/evidence/kotetes2010_mft.py`.

Verdict: PARTIAL — Coverage 6/10, Agreement 7/10.